

GRAND RAPIDS FORD, MI, USA

WMO: 726350

Lat: 42.883N

Lon: 85.524W

Elev: 803

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94860

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	6.8	-5.0	4.3	4.1	-0.4	5.5	8.2	27.2	27.0	25.0	26.3	8.2	200	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	89.4	72.9	86.7	71.7	84.2	70.2	76.3	85.2	74.4	82.8	72.8	80.7	12.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	127.9	81.2	71.6	120.4	79.1	70.1	113.9	77.5	40.2	85.4	38.4	82.7	36.8	80.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.3	19.1	DB	-6.0	93.9	6.3	3.2	-10.6	96.2	-14.3	98.0	-17.8	99.8	-22.4	102.1	(4)
(5)			WB	-6.6	78.8	6.2	2.2	-11.0	80.4	-14.6	81.7	-18.1	83.0	-22.6	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	24.2	26.2	35.3	47.4	58.8	68.6	72.4	70.7	63.6	51.4	39.7	30.0	(6)	
	DBStd	19.07	10.43	10.22	11.07	9.16	8.59	6.71	5.64	5.34	7.46	8.44	8.80	9.09	(7)	
	HDD50	3151	801	667	477	154	18	0	0	0	2	85	326	621	(8)	
	HDD65	6486	1266	1086	923	532	234	39	7	14	114	429	758	1084	(9)	
	CDD50	2839	0	1	21	76	290	557	695	640	410	129	18	2	(10)	
	CDD65	698	0	0	2	4	41	146	236	189	72	8	0	0	(11)	
	CDH74	6004	0	0	13	51	404	1315	2143	1423	588	67	0	0	(12)	
	CDH80	1753	0	0	2	7	104	407	710	366	149	8	0	0	(13)	
(14) Wind	WSAvg	9.3	10.6	10.6	10.3	10.9	9.5	8.3	7.8	7.2	7.7	9.1	9.9	10.2	(14)	
(15) Precipitation	PrecAvg	37.1	2.2	1.7	2.5	3.7	3.5	3.7	3.5	3.5	3.8	3.3	3.2	2.6	(15)	
	PrecMax	47.5	4.7	4.6	5.1	11.1	10.0	8.9	8.8	8.5	11.9	9.7	7.9	6.6	(16)	
	PrecMin	22.8	0.5	0.3	0.6	0.8	0.9	0.2	0.8	0.2	0.0	0.6	0.5	0.7	(17)	
	PrecStd	5.9	1.1	1.0	1.2	1.6	2.0	1.9	1.9	2.0	2.5	1.9	1.7	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	58.0	74.1	80.3	87.5	91.5	94.5	90.8	88.9	80.4	68.2	58.5	(19)	
		MCWB	52.7	52.0	61.6	62.6	69.0	73.9	76.5	74.6	70.4	67.9	58.0	54.7	(20)	
	2%	DB	47.5	50.0	65.3	73.7	83.0	88.1	89.8	87.0	84.2	75.0	61.9	51.8	(21)	
		MCWB	44.5	45.0	55.5	59.0	67.0	71.6	74.3	72.4	69.8	63.6	54.9	48.2	(22)	
	5%	DB	41.8	43.9	58.0	68.4	78.5	84.8	86.8	84.0	80.5	70.2	57.9	46.7	(23)	
		MCWB	38.9	39.5	50.8	55.3	64.5	70.2	72.5	70.6	67.7	61.7	51.8	43.2	(24)	
	10%	DB	37.7	39.5	51.8	63.4	74.2	81.6	83.9	81.3	76.9	65.8	53.4	41.7	(25)	
		MCWB	35.7	36.2	45.3	52.9	62.5	68.6	70.6	69.3	66.0	58.7	48.1	38.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	53.5	63.1	66.0	72.5	77.1	79.5	77.7	75.3	70.3	60.0	55.4	(27)	
		MCDB	54.7	56.0	73.3	76.7	83.4	87.2	88.6	87.0	83.8	78.3	64.6	57.5	(28)	
	2%	WB	45.1	45.4	57.6	61.1	69.3	74.2	77.1	75.1	72.5	66.5	56.4	49.3	(29)	
		MCDB	47.1	49.7	63.3	69.4	78.6	83.9	86.3	82.6	79.7	71.3	60.7	51.7	(30)	
	5%	WB	39.3	40.1	51.7	57.7	67.0	72.4	75.0	73.3	70.1	62.9	52.5	43.6	(31)	
		MCDB	41.2	43.4	56.9	66.0	75.4	81.8	83.9	80.1	77.0	68.8	57.1	46.1	(32)	
	10%	WB	35.5	36.3	45.7	53.7	64.2	70.5	73.1	71.8	67.9	59.3	48.4	38.8	(33)	
		MCDB	37.8	39.2	51.3	62.6	72.4	78.8	81.0	78.6	75.1	64.9	53.0	41.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	13.2	16.6	19.7	19.8	19.4	19.3	18.4	19.6	17.3	13.6	10.6	(35)	
		MCDBR	16.1	17.5	24.1	27.0	24.1	22.4	21.6	20.7	23.0	23.7	19.6	16.6	(36)	
	5% WB	MCWBR	14.3	14.3	16.2	15.4	12.4	10.3	9.8	9.5	10.9	14.3	14.3	14.1	(37)	
		MCWBR	14.9	16.5	22.0	23.3	20.8	18.9	18.8	17.0	18.7	20.7	17.8	15.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.296	0.346	0.342	0.378	0.411	0.431	0.421	0.422	0.384	0.348	0.316	0.295	(40)	
		taud	2.348	2.178	2.356	2.284	2.241	2.229	2.256	2.253	2.341	2.439	2.472	2.410	(41)	
	Ebn at Noon		266	265	285	282	274	268	269	265	267	265	258	256	(42)	
		Edh at Noon	26	36	35	40	43	44	42	41	35	28	23	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		413	682	1120	1466	1713	1931	1960	1673	1344	838	507	338	(44)	
	RadStd		37	90	119	139	170	125	91	95	126	95	70	42	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page